Which Half of Participation Does the Sparse-Averaging Contraction Need?

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Abstract

Decentralized training synchronizes model replicas by periodically averaging a random subset of their weights. Existing guarantees assume every replica both contributes to the average and receives the result, although these are separate communication events that fail independently in real systems. We audit this assumption and find an asymmetry: send failures do not affect the leading-order contraction of disagreement, and replicas agree exactly when every replica receives, however few contribute; receive failures reduce the contraction linearly. Send failures instead displace the value replicas agree on, a second channel that doubly stochastic mixing cannot represent because it preserves the mean by construction. The separation gives an effective averaging rate set by the receive fraction, showing the admissible step size is smaller than previously stated, and reveals staleness correction as a finite-time transfer of error between the two channels. All expressions are exact for arbitrary replica weights and validated against a faithful implementation without fitted parameters.

1 One Assumption, Two Events

Setting. m replicas hold weights w_i and take local optimizer steps. Periodically a random p-fraction of coordinates is selected, and for each the replicas exchange values and adopt the mean. With disagreement $\|\Delta\|^2 = \sum_i \|w_i - \bar{w}\|^2$, the standard guarantee is that one round contracts it to $(1-p)\|\Delta\|^2$ [Beton et al., 2025, Ajanthan et al., 2026].

Where the assumption hides. That proof sets $w_{i:\mu} \leftarrow \frac{1}{m}\sum_j w_{j:\mu}$ for every selected coordinate μ. The left-hand side carries the replica index i; the right-hand side does not. Every replica is assumed both to **contribute** to the mean and to **adopt** it. In deployment these are two network events with independent failure modes, instrumented separately: a permissionless 8.6B-parameter run logs "reduce success" and "averaging success" as distinct counters [Avraham et al., 2026]. No analysis in the sparse-averaging line separates them; the closest treatment is federated learning over noisy channels, which finds downlink noise more damaging than uplink [Wei and Shen, 2021, Upadhyay and Hashemi, 2023], the analogue for channel noise of what we prove here for participation.

The audit. Let φ_{g} be the fraction of replicas whose values reach the aggregator, and φ_{s} the fraction that receive the result back, drawn independently; $\varphi_{\mathrm{g}} = \varphi_{\mathrm{s}} = 1$ recovers the standard operator. We compute one round exactly for an **arbitrary fixed** weight configuration, with a constant independent of it (Appendix A): no distributional assumption and no replica homogeneity, where the derivation it extends assumes identical initialisation, i.i.d. splits and identical hyperparameters [Ajanthan et al., 2026]. Ours therefore covers heterogeneous data, but is one round, not a convergence guarantee under heterogeneous objectives. Per averaged coordinate, disagreement retained is $(1-\varphi_{\mathrm{s}})+O(1/m)$, so φ_{g} is absent at leading order. The corresponding shift in the mean is $\left[\varphi_{\mathrm{s}}^2(1-\varphi_{\mathrm{g}})/\varphi_{\mathrm{g}}+\varphi_{\mathrm{s}}(1-\varphi_{\mathrm{s}})\right]$ disagreement$/(m-1)$, where φ_{g} appears explicitly and the displacement diverges as $\varphi_{\mathrm{g}} \to 0$.

Why existing analyses cannot express this. The standard framework models a round as multiplication by a mixing matrix W and bounds disagreement through its spectral gap [Koloskova et al.,

2020]; recent systems work models failure as *weaker mixing* under a W taken "symmetric and doubly stochastic" [Hewa Koneputugodage et al., 2026]. Double stochasticity is exactly $\mathbf{1}^\top W = \mathbf{1}^\top$, so the round preserves the mean: the send channel is not merely unstudied there, it is zero by assumption, and no contraction factor can express it. Our operator loses double stochasticity as soon as the two sets differ, since columns of non-contributing replicas sum below one. The remedy for such mixing is push-sum [Kempe et al., 2003, Nedić and Olshevsky, 2016, Assran et al., 2019], which appears in no sparse-averaging system in this line. Non-mean-preserving consensus is itself classical [Fagnani and Zampieri, 2008]; what is new is its composition with sparse coordinate masking and local steps, the exact constants of both channels, and the consequence for the step size (Appendix I). The boundary is sharp: when the two sets coincide ($G = R$) the operator *is* doubly stochastic, retained disagreement is $m(1-\phi)/(m-1)$ and displacement is exactly zero, below 10^{-32} (Appendix G). Correlated failure is covered by existing theory, independent failure is not, and deployments lie between.

The contraction therefore needs only the receive half. Send failures are close to free for agreement and instead charge into a second channel that displaces the value agreed upon, shrinking as $1/m$. The $\varphi_s{=}1$ corner alone gives none of this: not the rate for $\varphi_s < 1$, not the displacement variance, and not the *opposing* motion of the two channels as failures are reallocated (Table 1).

2 What the Audit Buys

The admissible step size is smaller than stated. Composing the contraction with a standard local-SGD recursion gives, for L-smooth objectives, $\eta < p\,\varphi_s/[c(1-p\,\varphi_s)L]$ with an absolute constant c that also corrects the published $\eta < p/[2(1-p)L]$ (Appendix B), whose tightness Ajanthan et al. [2026] leave open. At fixed c, $\varphi_s{=}1$ gives $\eta < p/[c(1-p)L]$, so receive failure alone *narrows* the range. The effective sparsity is $p\,\varphi_s$, not p: a run losing a quarter of its returns averages 3.75% of coordinates rather than 5%, so a step size tuned on the nominal p exceeds this sufficient condition. Both channels enter as floors rather than growth rates, since the displacement is conditionally zero-mean and acts as additive noise, so within this analysis it raises the floor rather than acting at first order (Appendix B).

What a staleness corrector acts on. The split says what a delay corrector is for, and yields a design consequence: correctors and receive-path reliability act on the same channel and should be tuned jointly, not independently. Delay leaves the contraction nearly unchanged and increases displacement, so it acts on the second channel; the EMA corrector of Ajanthan et al. [2026] restores that displacement toward its delay-free value and increases the disagreement left standing (Appendix C, Table 3). For the corrector and delays tested, correction moves error between channels rather than removing it, and is least attractive where receive-path failure has already loaded the contraction channel.

This is not an artifact of operating near a known instability boundary: over two orders of magnitude of delay the residue grows monotonically with $\Pi = \tau p$, the stale writes in flight, from 0.015 to 0.401 as Π goes 0.05 to 1, while uncorrected averaging annihilates disagreement exactly at every delay when $\varphi_s{=}1$ (Appendix C, Table 4). This does not contradict the correction's asymptotic guarantee, which shows the spread vanishes and identifies the post-write deviation as exactly it; we measure the same quantity at finite time and show its value is set by Π, which the asymptotic statement does not expose. Per seed the budget sweep is monotone across all five splits, and three independent five-point orderings agreeing by chance has probability $\approx 5 \times 10^{-6}$.

Table 1: A fixed 25% failure budget moved from the receive to the send path ($m{=}16$, 3 seeds). $\psi{=}\varphi_s{+}O(1/m)$: effective receive fraction. Loss follows ψ; the predicted displacement moves the opposite way, 0.0125 to 0.0222.

send/recv	ψ	loss
0.00/0.25	0.763	4.0584
0.06/0.19	0.822	4.0521
0.13/0.13	0.881	4.0449
0.19/0.06	0.941	4.0382
0.25/0.00	1.000	4.0328

The operator behaves as derived. On real training weights across 11 configurations with both channels injected, prediction and measurement agree to 3.2×10^{-3} with no fitted constant (Appendix C, Table 2). A pre-registered prediction that the dominant failure path would swap with replica count is **falsified** at every tested m, including $m{=}4$ where the send channel is strongest (Appendix E), so the rule carries no m-dependent caveat.

The rule (at the scale tested; Appendix J). Spend reliability budget on making replicas *receive* rather than send, carry $p\,\varphi_s$ in guarantees rather than p, and report a staleness corrector with the disagreement it buys back.

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A The operator, and the two theorems

Notation. φ_{g} is the send (contribute) fraction and φ_{s} the receive (adopt) fraction, exactly as in the main text; the appendices use the same two symbols throughout. ψ, which appears below and in Table 1, is the *effective* receive fraction after the finite-population correction,

$$\psi \;=\; \varphi_{\mathrm{s}} - \frac{\varphi_{\mathrm{s}}(1-\varphi_{\mathrm{s}})}{m-1}\Big[\frac{1-\varphi_{\mathrm{g}}}{\varphi_{\mathrm{g}}} - 1\Big] \;=\; \varphi_{\mathrm{s}} + O(1/m),$$

i.e. what replaces φ_{s} when the contraction is written over the whole model rather than per coordinate. The symbol σ_g below denotes the residual sum $\sum_{i\notin R}\delta_i$ and is unrelated to the send fraction.

m replicas hold weights $w_i \in \mathbb{R}^d$ and take local optimizer steps between averaging rounds. Let $\bar{w} = \frac{1}{m}\sum_i w_i$ and let the consensus error be $\|\Delta\|^2 = \sum_i \|w_i - \bar{w}\|^2$.

Each round a mask selects every coordinate independently with probability p. Fix one selected coordinate and let $\delta_i = w_i - \bar{w}$ be the replica deviations at it, so $\sum_i \delta_i = 0$; write $S = \sum_i \delta_i^2$.

Complete operator (what theory assumes). Every replica sets $w_i \leftarrow \frac{1}{m}\sum_j w_j$. Disagreement at that coordinate becomes zero, giving $\mathbb{E}\|\Delta'\|^2 = (1-p)\|\Delta\|^2$ [Ajanthan et al., 2026].

Incomplete operator (what systems run). Let G with $|G| = \varphi_{\mathrm{g}} m$ be the set that reaches the reducer and R with $|R| = \varphi_{\mathrm{s}} m$ the set that receives the result, drawn uniformly and independently. The reducer forms

$$a \;=\; \frac{1}{|G|}\sum_{i\in G} w_i \;=\; \bar{w} + b, \qquad b \;:=\; \frac{1}{|G|}\sum_{i\in G}\delta_i, \tag{1}$$

and the write is $w_i' = a$ for $i \in R$, $w_i' = w_i$ otherwise. Setting $\varphi_{\mathrm{g}} = \varphi_{\mathrm{s}} = 1$ recovers the complete operator.

Why this is general. Everything below is exact for an *arbitrary fixed* δ; only G and R are random. That is far weaker than the homogeneous setup of Ajanthan et al. [2026], identical initialization, i.i.d. splits, identical hyperparameters, and data heterogeneity does not touch it.

Both proofs use the finite-population identities for a uniform $A \subseteq [m]$, $|A| = k$, over the fixed population $\{\delta_i\}$:

$$\mathbb{E}\Big[\sum_{i\in A}\delta_i\Big] = 0, \quad \mathbb{E}\Big[\big(\sum_{i\in A}\delta_i\big)^2\Big] = \frac{k(m-k)}{m-1}\frac{S}{m}, \quad \mathbb{E}\Big[\sum_{i\in A}\delta_i^2\Big] = \frac{k}{m}S. \tag{2}$$

Theorem 1 (The receive path sets the contraction)**.** *With $S' = \sum_i (w_i' - \bar{w}')^2$ and $\bar{w}' = \frac{1}{m}\sum_i w_i'$,*

$$\frac{\mathbb{E}[S']}{S} \;=\; (1-\varphi_{\mathrm{s}}) \;+\; \frac{\varphi_{\mathrm{s}}(1-\varphi_{\mathrm{s}})}{m-1}\Big[\frac{1-\varphi_{\mathrm{g}}}{\varphi_{\mathrm{g}}} - 1\Big]. \tag{3}$$

Over the whole model, $\mathbb{E}\|\Delta'\|^2 \;=\; (1 - p\,\psi)\|\Delta\|^2$, with ψ the effective receive fraction defined above.

Proof. Since $\bar{w}' - \bar{w} = (|R|b + \sigma_g)/m$ with $\sigma_g := \sum_{i\notin R}\delta_i$,

$$S' \;=\; \sum_i (w_i' - \bar{w})^2 - m(\bar{w}' - \bar{w})^2 \;=\; |R|\,b^2 + \sum_{i\notin R}\delta_i^2 - \frac{(|R|b+\sigma_g)^2}{m}.$$

Take $\mathbb{E}_G$ with R, δ fixed. By (2), $\mathbb{E}_G[b] = 0$ and $\mathbb{E}_G[b^2] = \frac{(m-|G|)S}{|G|m(m-1)}$, so $\mathbb{E}_G[(|R|b+\sigma_g)^2] = |R|^2\mathbb{E}_G[b^2] + \sigma_g^2$ and $\mathbb{E}_G[S'] = |R|(1-\frac{|R|}{m})\mathbb{E}_G[b^2] + \sum_{i\notin R}\delta_i^2 - \frac{\sigma_g^2}{m}$. Now take $\mathbb{E}_R$, again by (2): $\mathbb{E}_R[\sum_{i\notin R}\delta_i^2] = \frac{m-|R|}{m}S$ and $\mathbb{E}_R[\sigma_g^2] = \frac{(m-|R|)|R|}{m-1}\frac{S}{m}$. Substituting $|R| = \varphi_{\mathrm{s}} m$ and $|G| = \varphi_{\mathrm{g}} m$ gives $\frac{\varphi_{\mathrm{s}}(1-\varphi_{\mathrm{s}})(1-\varphi_{\mathrm{g}})}{\varphi_{\mathrm{g}}(m-1)}S + (1-\varphi_{\mathrm{s}})S - \frac{\varphi_{\mathrm{s}}(1-\varphi_{\mathrm{s}})}{m-1}S$, which is (3). □

Corollary 2. (i) $\varphi_{\mathrm{g}}{=}\varphi_{\mathrm{s}}{=}1$ *gives* $\mathbb{E}[S'] = 0$*: the complete operator's factor* $(1-p)$. (ii) $\psi = \varphi_{\mathrm{s}} + O(1/m)$*, so the send path is absent from the consensus recursion to leading order.* (iii) $\varphi_{\mathrm{s}} = 1$ *gives* $\mathbb{E}[S'] = 0$ *for* any $\varphi_{\mathrm{g}} > 0$.

Reading (iii) is the surprise: at $\varphi_s = 1$ the replicas agree *exactly*, however few of them contributed. Away from that corner the statement is leading-order in $1/m$, as in (ii). Sending is therefore not what the contraction needs at leading order. So what does the send path cost?

Theorem 3 (The send path moves the mean)**.** *With $v := S/m$,*

$$\mathbb{E}\big[(\bar{w}' - \bar{w})^2\big] = \frac{v}{m-1}\Big[\varphi_s^2\,\frac{1-\varphi_g}{\varphi_g} + \varphi_s(1-\varphi_s)\Big]. \tag{4}$$

Proof. $\mathbb{E}[(\bar{w}' - \bar{w})^2] = (|R|^2\mathbb{E}_G[b^2] + \mathbb{E}_R[\sigma_g^2])/m^2$; substitute (2). □

The bracket holds the two channels side by side. The send term $\varphi_s^2(1-\varphi_g)/\varphi_g$ blows up as $\varphi_g \to 0$ and vanishes at $\varphi_g = 1$. The receive term $\varphi_s(1-\varphi_s)$ peaks at $\varphi_s = 1/2$ and vanishes at both ends. At full participation both vanish and the mean is exactly preserved, as it must be.

The prefactor is what makes this more than a curiosity: **mean displacement is $O(1/m)$, contraction is $O(1)$.** The two channels shrink at different rates as a deployment grows.

B Derivation of the two floors

We assume f is L-smooth and bounded below by $f^\star$, stochastic gradients are unbiased with variance at most σ^2, and the mask, gather set and scatter set are drawn independently of the gradients. Write $V_t = \mathbb{E}\|\Delta^t\|^2$ and $\Delta_f = f(\bar{w}^0) - f^\star$.

Step 1: the consensus recursion. A local step expands the disagreement in the standard way [Yu et al., 2019]: $\mathbb{E}\|\hat{\Delta}^{t+1}\|^2 \le (1+2\eta L)V_t + \eta^2 C(m\sigma^2 + L^2 V_t)$, and Theorem 1 contracts it, giving

$$V_{t+1} \le (1-p\psi)\big[(1 + 2\eta L + \eta^2 C L^2)V_t + \eta^2 C m\sigma^2\big].$$

The recursion contracts when $(1-p\psi)\,(1+2\eta L+\eta^2 C L^2) < 1$, i.e. when $(1-p\psi)(2\eta L + C\eta^2 L^2) < p\psi$. The condition quoted in the main text of Ajanthan et al. [2026], and the form $\eta < p\psi/[2(1-p\psi)L]$ that replaces p by $p\psi$ in it, control the $2\eta L$ term alone and leave no room for $C\eta^2 L^2$, so neither is by itself sufficient. Since Step 3 already assumes $\eta \le 1/(2L)$, we have $C\eta^2 L^2 \le C\eta L/2$ and it suffices that

$$\eta \;<\; \frac{p\,\psi}{c\,(1-p\,\psi)\,L}, \qquad c = 2 + C/2, \tag{5}$$

which is the same condition with 2 replaced by the absolute constant c. Only the admissible range of η changes, by a constant factor; the floors derived below are unaffected, since they follow from the fixed point in the regime $\eta L \ll p\psi$. With (5) the fixed point satisfies

$$V_\infty \;\le\; \frac{(1-p\psi)\,\eta^2 C m\sigma^2}{1-(1-p\psi)(1+2\eta L+\eta^2 C L^2)} \;\asymp\; \frac{\eta^2 C m\sigma^2}{p\,\psi} \tag{6}$$

whenever $\eta L \ll p\psi$, which is the regime (5) already enforces: the denominator is $p\psi - (1-p\psi)(2\eta L + C\eta^2 L^2)$, which tends to $p\psi$ in that regime regardless of the constant c.

Step 2: the mean perturbation is conditionally zero-mean. Write $\bar{w}^{t+1} = \bar{w}^t - \eta\bar{g}^t + \xi^t$, where ξ^t collects the displacement of Theorem 3. Per coordinate $\xi = (|R|b + \sigma_g)/m$, and by (2) both $\mathbb{E}_G[b] = 0$ and $\mathbb{E}_R[\sigma_g] = 0$, so $\mathbb{E}[\xi^t \mid \mathcal{F}^t] = 0$. Consequently ξ^t contributes nothing to the inner-product term of the descent lemma and, being independent of $\bar{g}^t$, nothing to the cross term either. It enters only as $\frac{L}{2}\mathbb{E}\|\xi^t\|^2$, the mean-displacement term.

This is why, within this analysis, participation failure enters as a floor rather than as a drift. A biased perturbation would enter at first order and compound; a zero-mean one enters at second order and behaves as additive noise. This bounds the stationarity measure above by a quantity that does not grow with T; it is not a proof that no choice of step size or failure process can destabilise a run, and in particular it says nothing about step sizes outside (5).

Table 2: Measured one-round operator against Theorems 1 and 3, driving the real implementation on a training transformer (m=16, p=0.25). No fitted parameters. Max deviation 3.2×10^{-3}.

		disagreement retained		shift in the mean	
φ_g (send)	φ_s (receive)	measured	Thm 1	measured	Thm 3
1.00	1.00	0.00000	0.00000	0.00000	0.00000
0.75	1.00	0.00000	0.00000	0.02222	0.02222
0.50	1.00	0.00000	0.00000	0.06665	0.06667
1.00	0.75	0.23539	0.23750	0.01243	0.01250
1.00	0.50	0.48390	0.48333	0.01669	0.01667
0.75	0.75	0.23844	0.24167	0.02593	0.02500
0.50	0.50	0.49936	0.50000	0.03465	0.03333

Table 3: The same operator with a delay, and then with the EMA staleness correction of Ajanthan et al. [2026], at φ_g=1.0, φ_s=0.75, m=16. Predictions are the τ=0 values of Theorems 1–3; the departures are the point.

	consensus retained	shift in the mean
predicted ($\tau = 0$)	0.23750	0.01250
measured, plain write	0.23539	0.01243
measured, delay τ=10	0.27052	0.10237
measured, delay + EMA correction	0.73117	0.01298

Step 3: assemble. The descent lemma with $\|\frac{1}{m}\sum_i \nabla f(w_i) - \nabla f(\bar{w})\|^2 \le \frac{L^2}{m}\|\Delta\|^2$ gives, for $\eta \le 1/(2L)$,

$$\tfrac{\eta}{2}\mathbb{E}\|\nabla f(\bar{w}^t)\|^2 \;\le\; \mathbb{E}f(\bar{w}^t) - \mathbb{E}f(\bar{w}^{t+1}) + \tfrac{\eta L^2}{2m}V_t + \tfrac{L\eta^2\sigma^2}{2m} + \tfrac{L}{2}\Theta V_t.$$

Telescoping over T steps, dividing by $\eta/2$, and substituting (6) with $\psi = \varphi_s + O(1/m)$ yields the two floors: the $\frac{L^2}{m}V_\infty$ term becomes $L^2C\eta^2\sigma^2/(p\varphi_s)$, and $\frac{L}{\eta}\Theta V_\infty$ becomes $\frac{LC\eta\sigma^2}{m-1}\big[\varphi_s\frac{1-\varphi_g}{\varphi_g} + (1-\varphi_s)\big]$. □

C Does the operator behave as derived?

We drive the real sparse-averaging implementation with both failure channels injected and compare predicted against measured one-round quantities on training weights, across 11 configurations. There is no fitted constant: the predictions are Theorems 1 and 3 evaluated at the run's own $(\varphi_g, \varphi_s, m, p)$.

The seven rows above are a representative subset of the 11 configurations measured; the largest absolute deviation across all 11 is 3.2×10^{-3}. Note the first block: at φ_s=1 the disagreement retained is 0.00000 for every φ_g down to 0.50, while the shift in the mean grows monotonically, the two channels, separated, in one table.

Table 4: Disagreement left standing after one round, sweeping the delay at p=0.05, m=16, φ_g=φ_s=1. $\Pi = \tau p$ counts the stale writes to a coordinate in flight at once; two seeds per cell, agreeing to 5×10^{-4}. The Π=0.05 row is the regime of the deployed run of Avraham et al. [2026].

τ	Π	uncorrected	EMA corrected
1	0.05	0.00000	0.01508
5	0.25	0.00000	0.08715
10	0.50	0.00000	0.18929
20	1.00	0.00000	0.40132

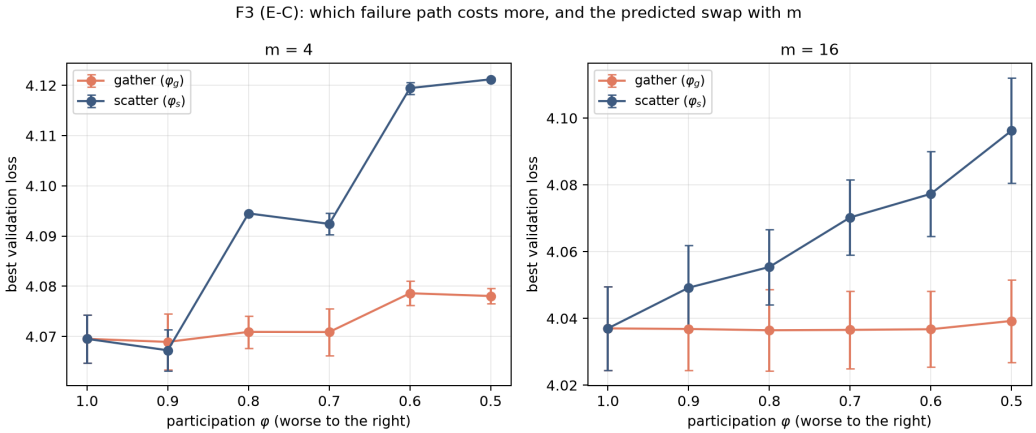


Figure 1: Validation loss against participation, sweeping each path separately, at $m{=}4$ and $m{=}16$, three seeds; error bars are half the seed range. The plotted legend predates the current notation: read *gather* (φ_{g}) as φ_{g} and *scatter* (φ_{s}) as φ_{s}. The prediction below is that the costlier curve swaps between panels. It does not.

D Correctors in the two-channel framework

Table 3 reports that a staleness corrector restores the mean and triples the consensus error. The decomposition explains why, in one line.

A delay-corrected write replaces the stale average with $w_i' = \bar{w}^{t-\tau} + d_i$, where d_i is the corrector's per-replica estimate of the drift. Decompose it as $d_i = \bar{d} + e_i$ with $\sum_i e_i = 0$. The written values now differ across replicas, so the post-write deviation on a selected coordinate is e_i rather than 0, and

$$\frac{\mathbb{E}[S']}{S} \;=\; \frac{\mathbb{E}\|e\|^2}{S}, \qquad \bar{w}' - \bar{w} \;=\; \big(\bar{w}^{t-\tau} + \bar{d}\big) - \bar{w}^{t}. \tag{7}$$

The corrector's *cross-replica spread* lands entirely in the contraction channel, and its *mean* entirely in the displacement channel. A corrector is thus exactly a device for moving error from the second channel to the first. Table 3 measures that exchange: the mean is restored to within 4% of its delay-free value, and the contraction channel absorbs the cost.

This also identifies when a corrector is a bad trade: when the contraction channel already dominates the floor bound of Appendix B. By Theorem 1 that is precisely the regime created by receive-path failure, so correcting staleness and losing returns interact badly, and the two should not be tuned independently.

E The crossover, tested and falsified

Theorem 3's $1/(m{-}1)$ prefactor makes the send channel $O(1/m)$ while Theorem 1 is $O(1)$, so there should be a replica count below which the send path is the costlier one. It should therefore cost more at small m, where the $1/(m{-}1)$ prefactor of Theorem 3 is largest. We tested it directly and it is false over the range we can reach. At $m{=}4$, where the send channel is at its strongest, five times its value at $m{=}16$, the receive path is still the costlier one, by 0.0420 in validation loss against a maximum seed spread of 0.0049. At $m{=}16$ the receive path is costlier by 0.0488, against a seed spread of 0.0315.

We report this as a negative result rather than removing the prediction, because it sharpens the practical claim rather than weakening it. The send channel is real, Theorem 3 is verified to $3.2{\times}10^{-3}$ in Table 2, but at realistic learning rates it sits beneath the optimization noise floor even where it is largest. The consequence is that the design rule stated in Section 2 carries no m-dependent caveat.

F The reliability-budget sweep

Holding the *total* failure rate fixed at 25% and varying only which path absorbs it isolates the asymmetry, and shows the two channels moving in *opposite* directions, the effect no reading of the $\varphi_{\mathrm{s}}{=}1$ corner predicts, and the end-to-end verification referred to in Section 1. Validation loss falls monotonically as the budget moves off the receive path onto the send path, tracking the predicted contraction rate ψ at all five splits, while the predicted mean displacement rises along the same sweep from 0.0125 to 0.0222. Per seed the all-receive minus all-send difference is $+0.0232$, $+0.0261$ and $+0.0274$; each seed is monotone across all five splits, and three independent five-point orderings agreeing by chance has probability $\approx 5 \times 10^{-6}$.

The measurements are Table 1 in the main text.

G Perfectly correlated failure ($G = R$)

Failures are drawn independently in the main analysis, while real ones correlate across the two phases. The perfectly correlated extreme is exactly solvable and brackets the independent case.

Let the contributing and adopting sets coincide, $G = R$, with $|G| = \phi m$. The aggregate is $a = \bar{w} + b$ with $b = \frac{1}{|G|}\sum_{i \in G} \delta_i$, and the write is $w_i' = a$ for $i \in G$ and $w_i' = w_i$ otherwise. Then

$$\bar{w}' - \bar{w} \;=\; \frac{1}{m}\Big[|G|\, b + \sum_{i \notin G} \delta_i\Big] \;=\; 0,$$

since $\sum_{i \notin G} \delta_i = -\sum_{i \in G} \delta_i = -|G|b$. The mean is preserved identically: with $G = R$ the operator is doubly stochastic and the send channel vanishes. This is the precise sense in which the mixing framework of Koloskova et al. [2020] and Hewa Koneputugodage et al. [2026] is exactly right for correlated failure and structurally blind to the independent case.

For the disagreement, $S' = |G|b^2 + \sum_{i \notin G} \delta_i^2$; taking the expectation over a uniform subset of size $|G|$ and using the finite-population variance of a zero-sum population gives $\mathbb{E}[S']/S = m(1-\phi)/(m-1)$, which is $(1-\phi)$ to leading order, the same rate the receive channel sets in the independent case, as it must be since every adopting replica is also a contributor here.

Table 5: Perfectly correlated failure, 4×10^5 draws per cell. The displacement is zero to machine precision and the retained disagreement matches the closed form.

m	ϕ	retained (measured)	$m(1-\phi)/(m-1)$	displacement
16	0.75	0.26638	0.26667	1.3×10^{-33}
16	0.50	0.53392	0.53333	1.3×10^{-33}
4	0.75	0.33298	0.33333	3.2×10^{-33}
4	0.50	0.66672	0.66667	3.0×10^{-33}

H Stall and recovery

The floor bound of Appendix B predicts that participation failure raises a floor rather than destabilising the recursion, so a run should stall while failures are active and recover when they stop. We switch failures on over steps $[2000, 4000)$ and release them, against two controls: a clean run, and two configurations where the delay corrector is unstable at *full* participation.

The separation is complete. All three participation runs, $\varphi_{\mathrm{s}} = 0.6$, $\varphi_{\mathrm{s}} = 0.4$, and $\varphi_{\mathrm{g}} = 0.6$, pass the divergence criterion, as does the clean control. Both delay-corrector configurations diverge and do not recover. Participation failure and staleness failure are therefore not two degrees of one problem: one moves a floor, the other breaks a contraction, and only the first is survivable.

I Related work

Sparse parameter averaging [Beton et al., 2025] and its variants [Fournier et al., 2024] exchange random weight subsets; Lee et al. [2023] analyse partial averaging with pre-determined subsets in

federated learning, where a similar learning-rate/subset-size relationship appears. Ajanthan et al. [2026] make sparse averaging asynchronous and add the EMA staleness correction priced here, while Douillard et al. [2023] reduce communication by averaging *infrequently* rather than sparsely. Maziane et al. [2026] give the first guarantee combining local steps, sparsification and overlap, and Maranjyan and Richtárik [2026] obtain optimal time complexity under heterogeneity, though through a server and assuming free communication. Every one of these analyses assumes the complete, common-mask operator: all replicas contribute, and all adopt.

That asymmetric link failure can preserve agreement while moving the value agreed upon is classical in randomised gossip [Fagnani and Zampieri, 2008], and we claim no novelty for the phenomenon. Our contribution is to locate that assumption inside a specific published contraction, to give the exact constants of both channels under sparse masking with local steps, and to show what the split implies for the admissible step size and for staleness correction.

J Experimental configuration

All runs use the 45M `small` preset (6 layers, d=512, block 512, GPT-2 tokenizer, tied embeddings) on WikiText-103, batch 8, AdamW at 3×10^{-4} with 600 warmup steps and cosine decay, weight decay 0.01, gradient clipping 1.0, bf16 autocast. The m replicas are stepped sequentially in a single process, which makes a run bit-reproducible given its seed, a requirement for the operator-exactness test of Appendix C, where predicted and measured quantities are compared on the same weights.

Sparse averaging uses the partitioned index selector of Beton et al. [2025], which splits each parameter into $\lceil 1/p \rceil$ disjoint parts and cycles them, so the realised fraction is $1/\lceil 1/p \rceil$ rather than p; every p we quote is a realised fraction. Failures are injected per replica per round, independently on the two phases, at rates φ_{g} and φ_{s}.

Relation to the sampling assumptions of Appendix A. Two differences between the analysis and the implementation are deliberate. First, Appendix A places each coordinate in the mask independently with probability p, whereas the partitioned selector cycles disjoint parts. Theorems 1 and 3 are stated *per averaged coordinate* and the operator acts on each selected coordinate separately and identically, so the law of the mask enters only through the realised fraction of coordinates averaged, which is the p we quote; no step of either proof uses independence across coordinates. Second, Appendix A fixes $|G| = \varphi_{\mathrm{g}} m$ and $|R| = \varphi_{\mathrm{s}} m$, whereas failures are injected per replica, so the realised cardinalities are Binomial rather than fixed and $1/\varphi_{\mathrm{g}}$ is convex. We therefore evaluate every prediction at the cardinalities a round actually realised — logged per round, and reported as the realised φ_{g} and φ_{s} — rather than at their nominal means, which is what makes the comparison in Appendix C a test of the identity rather than of a Jensen gap. Rounds in which no replica contributes to a coordinate are a no-op at that coordinate: no average exists, and the implementation masks the coordinate out so that no replica adopts anything. The $\varphi_{\mathrm{g}} \to 0$ divergence of Theorem 3 is therefore never evaluated at $|G| = 0$; it describes rounds with few contributors, not none.

Grids, seeds and the exact argument string for every run are generated programmatically and recorded as run manifests, and the run-level JSON summaries are keyed by the same names, so every number here is traceable to one command. Divergence is decided by a pre-registered rule fixed before any run (non-finite loss; validation loss at or above $\ln V$ after step 500; or validation loss regressing more than 0.5 above its best), never by inspection. Code, the per-run JSON summaries behind every table, and the camera-ready source are at `https://github.com/Pusse-01/sparse-averaging-particpation`.